

Case Spam Classification Using Gen AI (OpenAI LLM)

Problem Statement:

Customer service teams, email marketers, and IT administrators often spend significant time reviewing email content to detect spam. This manual process can be inefficient, error-prone, and resource-intensive. Despite employing traditional spam filters, they struggle with false positives and missed spam, leading to either important emails being incorrectly flagged or harmful content infiltrating inboxes. As email volumes increase, the complexity of detecting spam through human intervention becomes unsustainable, resulting in potential security risks and inefficiencies in communication.

Goal:

The objective of this work is to demonstrate how Generative AI (like OpenAI's GPT models) can be applied in email systems to automate spam classification. Accurate spam detection ensures that spam emails are effectively filtered, improving the user experience, protecting systems from potential threats, and reducing manual intervention.

Approach:

Generative AI models can interpret email subject lines and descriptions to automatically assess:

- **Spam likelihood** (e.g., Spam or Not Spam),
- **Content-based triggers** (e.g., Suspicious links, Offers, Scams),
- **Email characteristics** (e.g., Unusual requests, urgency, promotional language).

By using an AI-driven approach, we can automate spam detection by analyzing unstructured email content to derive insights.

Use Cases

50 different email conversations of different combinations of subject & description emails cases are given to LLM and output is documented. The results look very promising with Few Shot prompt

Here is the prompt Template

```
"""
    Analyze the email subject and description using OpenAI GPT to determine
    if it is spam or not.
    """
    prompt = f"""
    You are an email classifier. Determine if the following email is spam
    based on its subject and description.
    Provide a classification: 'Spam' or 'Not Spam' with a short
    explanation.

    Subject: {subject}
    Description: {description}
    """
    """
```

	Subject	Description	LLM output Spam/Not Spam	LLM reason
1	Subject: Monthly Newsletter from XYZ Corp	Description: We hope you're enjoying our latest updates. Check out what's new this month in your industry.	Not Spam	Explanation: The email has a subject related to a monthly newsletter from XYZ Corp and the description mentions updates in the recipient's industry. This content is typical of a legitimate newsletter and does not exhibit any characteristics commonly associated with spam emails.

2	Subject: Your Invoice #123456 from ABC Corp	Description: Please find your invoice for the recent services you availed from us. Let us know if you have any questions.	Not Spam	Explanation: This email appears to be a legitimate communication regarding an invoice from ABC Corp. The subject and description are related to a specific transaction, and there is no indication of typical spam characteristics such as unsolicited offers or suspicious links.
3	Subject: Invitation to Join Webinar on Data Privacy	Description: Join us for a live session on upcoming data privacy regulations and how your business can prepare.	Not Spam	Explanation: The email is inviting the recipient to join a webinar on data privacy, which seems relevant and legitimate. There are no typical spam indicators such as suspicious links, offers, or requests for personal information.
4	Subject: Confirmation of Your Flight Booking	Description: Your flight from New York to London has been confirmed. Find attached your e-ticket and itinerary.	Not Spam	Explanation: The email appears to be a legitimate confirmation of a flight booking from New York to London, providing the recipient with their e-ticket and itinerary. There are no typical spam indicators present in the subject or description, so it is classified as Not Spam.
5	Subject: Job Interview Scheduled for Tomorrow	Description: We are excited to meet you. Your interview is scheduled for 10:00 AM tomorrow. Best of luck!	Not Spam	Explanation: This email appears to be a legitimate communication regarding a job interview scheduled for tomorrow. The subject and description are relevant and do not contain typical characteristics of spam emails.
6	Subject: You Won \$1,000,000 - Claim Your Prize Now!	Description: Click here to claim your grand prize. This offer is only valid for 24 hours!	Spam	Explanation: The subject line and description of the email suggest that it is a typical spam message promising a large sum of money as a prize, with urgency to claim it within a short timeframe. These characteristics are common in spam emails designed to lure recipients into clicking on malicious links or providing personal information. It is

				advisable to be cautious and avoid interacting with such emails.
7	Subject: Earn \$500 Daily From Home Without Any Experience!	Description: Start earning today with our easy online job. No skills required. Sign up now!	Spam	Explanation: The email subject and description contain typical spam characteristics such as promises of easy money without any experience or skills required. This is a common tactic used by spam emails to lure recipients into potential scams or fraudulent schemes.
8	Subject: Urgent: Your Account Has Been Suspended	Description: We have detected suspicious activity in your account. Click this link to resolve the issue immediately.	Spam	Explanation: The email subject implies urgency and fear by stating that the recipient's account has been suspended. Additionally, the description prompts the recipient to click on a link to resolve the issue, which is a common tactic used in phishing emails. These characteristics are typical of spam emails attempting to deceive recipients into providing personal information or clicking on malicious links.
9	Subject: Limited Time Offer: Free Vacation to the Bahamas	Description: You've been selected for a free vacation! All you need to do is cover the processing fee.	Spam	Explanation: The email is likely spam because it offers a free vacation to the Bahamas but requires the recipient to cover a processing fee. This is a common tactic used by scammers to lure people into paying money for something that may not actually exist or be as advertised. It's always advisable to be cautious of such offers, especially when they involve upfront payments.
10	Subject: You Have Been Selected for a \$100 Amazon Gift Card	Description: Congratulations! Click here to claim your free Amazon gift card.	Spam	Explanation: The email subject and description are typical characteristics of spam emails, promising a free gift card in exchange for clicking a link. This is a common tactic used by spammers to lure recipients into

				providing personal information or falling victim to scams. It is best to avoid interacting with such emails to protect your personal information and security.
1	Subject: Project Update: Q4 Strategy Meeting	Description: Please find attached the latest updates from our Q4 planning session. Let me know your thoughts.	Not Spam	Explanation: The email has a subject related to a project update and the description mentions sharing updates from a Q4 planning session. This content appears to be work-related and does not exhibit typical characteristics of spam emails.
2	Subject: Friendly Reminder: Team Lunch Tomorrow	Description: Just a reminder that we have our team lunch at 12 PM tomorrow. Don't forget to RSVP!	Not Spam	Explanation: This email appears to be a genuine reminder about a team lunch event. The subject and description are related to a work-related activity, and there is no indication of any spammy content or malicious intent.
3	Subject: Your Amazon Order Has Shipped	Description: Your recent order has been shipped and is expected to arrive by Friday. Track your package here.	Not Spam	Explanation: The email appears to be a legitimate notification about an Amazon order being shipped. It provides relevant information about the order status and includes a link to track the package, which is common in shipping confirmation emails from reputable companies like Amazon. Therefore, this email is classified as 'Not Spam'.
4	Subject: Invitation: Networking Event in Your Area	Description: You're invited to a networking event with industry leaders. Join us for an evening of insights and connections.	Not Spam	Explanation: The email is inviting the recipient to a networking event with industry leaders, which seems like a legitimate and professional opportunity. There are no obvious signs of spam such as unsolicited offers, suspicious links, or requests for personal information.
5	Subject: Weekly Status Update: Product Development	Description: Here's your weekly update on the product development progress. We're on track for the Q1 launch.	Not Spam	Explanation: The email has a legitimate subject related to a weekly status update on product development progress. There is no indication of spammy content

				or malicious intent in the description provided.
1 6	Subject: Your Loan Has Been Pre-Approved	Description: You are eligible for a loan up to \$100,000 with no credit check! Apply now to get instant approval.	Spam	Explanation: The email subject and description are typical characteristics of spam emails offering loans with no credit check and instant approval. These types of emails often target individuals with promises of easy money but may lead to scams or fraudulent activities. It is advisable to be cautious and avoid engaging with such unsolicited offers.
1 7	Subject: Immediate Action Required: Update Your Payment Information	Description: We couldn't process your recent payment. Click here to update your payment details immediately.	Spam	Explanation: The email subject and description are designed to create a sense of urgency and prompt the recipient to click on a link to update their payment information. This is a common tactic used by phishing emails to trick users into providing sensitive information. It is advisable to be cautious and not click on any links in such emails.
1 8	Subject: Congratulations! You've Been Chosen for a Special Offer	Description: You've been selected for a limited-time offer. Click now to avail this exclusive deal.	Spam	Explanation: The email subject and description contain typical characteristics of spam emails, such as claiming the recipient has been chosen for a special offer without specifying any details. Additionally, the urgency to click on a link for an exclusive deal is a common tactic used in spam emails to lure recipients into clicking on potentially malicious links.
1 9	Subject: Get 100,000 Followers Instantly on Social Media	Description: Boost your social media presence with 100,000 followers. Click here to get started now!	Spam	Explanation: The email is likely spam as it promises a large number of followers instantly, which is a common tactic used in spam emails to lure recipients into clicking on potentially harmful links or providing personal information. Additionally, the subject line and

				description use phrases often associated with spam emails.
20	Subject: Hot Singles in Your Area Looking to Chat	Description: Find singles near you! Join our platform today and start connecting instantly.	Spam	Explanation: The subject line and description are typical of spam emails that promote dating or adult content. The language used suggests a sense of urgency and the promise of connecting with singles, which are common tactics used in spam emails.
21	Subject: Your Subscription Renewal Reminder	Description: Your subscription to our premium service is set to renew in 5 days. No action is required unless you wish to cancel.	Not Spam	Explanation: This email appears to be a legitimate subscription renewal reminder for a premium service. It does not contain any typical spam characteristics such as unsolicited content, suspicious links, or requests for personal information.
22	Subject: Meeting Confirmation: Marketing Strategy Discussion	Description: This is a confirmation for your meeting with the marketing team on Monday at 2 PM.	Not Spam	Explanation: The email has a subject related to a meeting confirmation for a marketing strategy discussion, which indicates it is likely a legitimate business communication rather than spam.
23	Subject: Re: Follow-up on the Contract Details	Description: I'm following up on the contract we discussed last week. Let me know if you have any further questions.	Not Spam	Explanation: The email subject and description indicate a legitimate follow-up on a contract discussion, which is a common business communication. There are no typical spam indicators such as suspicious links, offers, or irrelevant content.
24	Subject: Welcome to the Team!	Description: We are thrilled to have you join our company. Looking forward to seeing you at the orientation next week.	Not Spam	Explanation: The email has a subject line "Welcome to the Team!" and a description that indicates the recipient has joined a company and is invited to an orientation next week. This content does not appear to be spam as it relates to welcoming a new team member and providing information about an upcoming event within the company.

2 5	Subject: Your Package is Out for Delivery	Description: Your order is on its way! Expect delivery by the end of the day.	Not Spam	Explanation: The email subject and description indicate that it is a legitimate notification about a package delivery. There are no typical spam characteristics present in the content provided.
2 6	Subject: Survey: Help Us Improve Our Services	Description: Please take 5 minutes to fill out this survey and help us provide better services for you.	Not Spam	Explanation: The email has a subject related to a survey aimed at improving services, which is a common practice for companies seeking feedback from customers. The description also clearly states the purpose of the survey and does not contain any suspicious or promotional content typically found in spam emails. Therefore, this email is classified as 'Not Spam'.
2 7	Subject: Invitation: Join Our Customer Success Webinar	Description: We'd like to invite you to a webinar focused on customer success strategies for 2024.	Not Spam	Explanation: The email is inviting the recipient to join a webinar focused on customer success strategies for 2024, which seems like a legitimate and relevant topic. There are no obvious signs of spam such as unsolicited offers, suspicious links, or irrelevant content.
2 8	Subject: Confirm Your Attendance at the Annual Gala	Description: We are excited to have you join us for our Annual Gala next month. Please confirm your attendance.	Not Spam	Explanation: The email has a legitimate subject related to confirming attendance at an Annual Gala event. There are no typical spam indicators present in the subject or description, so it is likely not spam.
2 9	Subject: Your Payment Has Been Received	Description: Thank you for your payment. Your transaction has been successfully processed.	Not Spam	Explanation: The email has a subject related to a payment being received and the description confirms that the transaction has been successfully processed. This content does not exhibit typical characteristics of spam emails, so it is classified as 'Not Spam'.

3 0	Subject: Action Required: Update Your Account Password	Description: As part of our security measures, we require you to update your password every 90 days.	Not Spam	Explanation: This email appears to be a legitimate security measure requiring the recipient to update their account password every 90 days. It does not exhibit typical characteristics of spam emails such as unsolicited offers, suspicious links, or deceptive content.
3 1	Subject: Get Rid of Your Debt Now!	Description: Struggling with debt? We can help you clear all your dues in no time! Click here to get started.	Spam	Explanation: The email subject and description are typical of spam emails that promise quick solutions to debt problems. The language used is often associated with deceptive or fraudulent practices. It's advisable to be cautious and avoid clicking on any links in such emails.
3 2	Subject: Click Here for Free Gift Cards!	Description: Get free gift cards from your favorite stores! Limited time offer, claim yours now.	Spam	Explanation: The email subject and description are typical of spam emails that promise free gift cards or rewards in exchange for clicking on a link. These types of emails often lead to scams or phishing attempts. It is advisable to be cautious and avoid clicking on such links.
3 3	Subject: Your Account Is About to Be Closed	Description: Your account will be deactivated unless you confirm your details. Click here to keep your account active.	Spam	Explanation: The email subject and description are typical characteristics of a phishing scam. It is trying to create a sense of urgency by claiming that the recipient's account is about to be closed unless they click on a link to confirm their details. This is a common tactic used by spammers to trick individuals into providing personal information or clicking on malicious links. It is advisable not to interact with such emails and report them as spam.

3 4	Subject: You've Won a Brand New Car!	Description: Congratulations! You've been selected to win a brand-new car. Just fill out this form to claim your prize.	Spam	Explanation: The email is likely spam because it is claiming that the recipient has won a brand new car without any prior knowledge or participation in a contest. Additionally, it is asking the recipient to fill out a form, which could potentially be used for phishing purposes.
3 5	Subject: Earn Passive Income from Home	Description: Start earning money with little effort. Join our program and make thousands every week!	Spam	Explanation: The subject and description of the email suggest that it is promoting a program to earn passive income from home, promising to make thousands every week with little effort. This kind of message often aligns with typical spam emails that aim to lure recipients into potential scams or schemes.
3 6	Subject: Last Chance to Claim Your Discount Voucher!	Description: This is your last chance to claim a 70% discount voucher. Don't miss out!	Spam	Explanation: The subject line and description of the email are typical characteristics of spam emails, often used to lure recipients into clicking on malicious links or providing personal information. The urgency and offer of a high discount percentage without any specific details make it likely to be a spam email.
3 7	Subject: Protect Your PC from Viruses Now	Description: Install our antivirus software and protect your PC from threats. Click here for a free trial.	Not Spam	Explanation: The email is promoting antivirus software to protect the recipient's PC from viruses, which is a common and legitimate service. There are no obvious red flags or deceptive tactics typically associated with spam emails.
3 8	Subject: Immediate Action Required: Resolve Payment Issue	Description: We were unable to process your recent payment. Click here to resolve the issue immediately.	Not Spam	Explanation: The email subject indicates that immediate action is required to resolve a payment issue, which is a common type of legitimate communication from companies or service providers. It does not exhibit typical characteristics of spam emails.

				such as unsolicited offers, suspicious links, or irrelevant content. Therefore, this email is classified as 'Not Spam'.
3 9	Subject: New Hot Deal: Get 70% Off on Electronics	Description: Save big on the latest gadgets. Hurry, offer ends soon!	Not Spam	Explanation: The email subject and description are promoting a discount offer on electronics, which is a common marketing tactic. There is no indication of malicious intent or fraudulent activity in the content provided. Therefore, this email is classified as 'Not Spam'.
4 0	Subject: Congratulations! You've Been Approved for a Credit Card	Description: Apply now and get approved for a high-limit credit card with no annual fee.	Spam	Explanation: The email is classified as spam because it is promoting a credit card without any prior interaction or consent from the recipient. Additionally, the subject line uses enticing language to lure the recipient into applying for a credit card, which is a common tactic used in spam emails.
4 1	Subject: Your Software License Has Been Renewed	Description: Your software license has been successfully renewed for the next 12 months.	Not Spam	Explanation: The email subject and description indicate that the software license has been renewed, which is a legitimate and expected communication from a software provider. There are no typical spam indicators such as unsolicited offers, suspicious links, or requests for personal information.
4 2	Subject: Client Feedback Request for Last Project	Description: Could you please provide feedback on the recent project we completed? We value your input.	Not Spam	Explanation: The email appears to be a genuine request for feedback on a recent project completed. There are no typical spam indicators such as suspicious links, offers, or requests for personal information. It seems like a legitimate communication seeking input from the recipient.

3	Subject: Travel Itinerary for Your Upcoming Trip	Description: Attached is the travel itinerary for your upcoming business trip. Let us know if you have any changes.	Not Spam	Explanation: The email appears to be a legitimate communication regarding a travel itinerary for an upcoming business trip. There is no indication of typical spam characteristics such as unsolicited content, suspicious links, or attempts to deceive the recipient.
4	Subject: Notification: New Features Added to Your Account	Description: We've added exciting new features to your account. Log in to explore the updates.	Not Spam	Explanation: The email subject and description indicate that new features have been added to the recipient's account, which is a common communication from legitimate service providers. There are no typical spam indicators such as urgent requests, suspicious links, or irrelevant content. Therefore, this email is classified as 'Not Spam'.
4	Subject: Reminder: Payment Due in 3 Days	Description: Your payment is due in 3 days. Please ensure timely payment to avoid any service interruptions.	Not Spam	Explanation: This email appears to be a legitimate reminder about an upcoming payment deadline. The subject and description are clear and relevant, indicating that it is not likely to be spam.
4	Subject: Weekly Newsletter: Industry Insights and News	Description: Here's this week's edition of our newsletter, filled with insights and news from your industry.	Not Spam	Explanation: The subject and description of the email indicate that it is a weekly newsletter providing industry insights and news, which suggests that it is likely a legitimate communication rather than spam.
4	Subject: Annual Company Retreat Details	Description: We are pleased to share the details for our upcoming annual company retreat. Please review the attached agenda.	Not Spam	Explanation: The email has a subject related to an "Annual Company Retreat Details" and the description mentions sharing details for the upcoming retreat along with an attached agenda. This content appears to be related to official company activities and does not exhibit typical characteristics of spam emails.

4 8	Subject: Job Offer: Congratulations, You're Hired!	Description: We are delighted to extend an offer for you to join our team. Please review the offer letter attached.	Not Spam	Explanation: The email appears to be a legitimate job offer, with the subject indicating that the recipient has been hired. There is a mention of reviewing an offer letter attached, which is a common practice in job offers. Based on this information, the email does not seem to be spam.
4 9	Subject: Appointment Confirmation: Dr. Smith at 10:30 AM	Description: This is a confirmation for your appointment with Dr. Smith tomorrow at 10:30 AM. Please arrive 15 minutes early.	Not Spam	Explanation: This email appears to be a legitimate appointment confirmation from Dr. Smith, providing specific details about the appointment time and instructions to arrive early. There are no typical spam indicators present in the subject or description, so it is classified as 'Not Spam'.
5 0	Subject: Upcoming Webinar: Boosting Sales with AI	Description: Join us for a webinar on how to boost your sales using AI-powered tools and techniques.	Not Spam	Explanation: The email is promoting an upcoming webinar on boosting sales with AI-powered tools and techniques. It does not exhibit any characteristics commonly found in spam emails such as unsolicited content, suspicious links, or deceptive tactics.

Google Colab Code

Link:

<https://colab.research.google.com/drive/1UiRFGg7FZn1QC0wirsajh4RGE9aGKAfq#scrollTo=aGjiP3Dpajbm>

```
!pip install openai
!pip install matplotlib

import matplotlib.pyplot as plt
from openai import OpenAI
```

```

import json
from google.colab import userdata

apikey = userdata.get('OPENAI_API_KEY')

# Set your OpenAI API key here
GPT_MODEL = "gpt-3.5-turbo"
client = OpenAI(api_key=apikey)

def analyze_email_for_spam(subject, description):
    """
    Analyze the email subject and description using OpenAI GPT to determine
    if it is spam or not.
    """
    prompt = f"""
    You are an email classifier. Determine if the following email is spam
    based on its subject and description.
    Provide a classification: 'Spam' or 'Not Spam' with a short
    explanation.

    Subject: {subject}
    Description: {description}
    """

    response = client.chat.completions.create(
        messages=[
            {'role': 'system', 'content': f'You are an email classifier'},
            {'role': 'user', 'content': prompt},
        ],
        model=GPT_MODEL,
        temperature=0.7,
        max_tokens=1000,
        top_p=0.5,
        n=1,
        stop=None,
        frequency_penalty=0.6,
        presence_penalty=0.8,
        stream=False,
        logprobs=False,
    )
    result = response.choices[0].message.content.strip()

    # Based on the analysis, assign a classification score

```

```

        classification, explanation = classify_spam(result)

    return classification, explanation

def classify_spam(analysis):
    """
    Classify the email as spam or not based on the analysis.
    """
    if "spam" in analysis.lower():
        classification = "Spam"
    else:
        classification = "Not Spam"

    explanation = analysis
    return classification, explanation

# Read input data from a text file
with open('sample_data/input.txt', 'r') as input_file:
    email_data = input_file.readlines()

# Open output file to write the results
with open('sample_data/output.txt', 'w') as output_file:
    for email in email_data:
        # Assuming the email data is in the format:
        "ID\tSubject\tDescription"
        email_id, subject, description = email.split('\t', 2)
        classification, explanation = analyze_email_for_spam(subject,
description)

        # Write the analysis and classification to the output file
        output_format = f"Email {email_id}: {classification} - Explanation:
{explanation}"
        output_file.write(output_format + "\n")
        print(output_format)

print("Spam detection processing completed. Results have been written to
email_output.txt.")

print("Processing completed. Results have been written to output.txt.")

```

Input.txt

The above code will read the input emails from this file (Name of file should be:input.txt)

```
1      Subject: Monthly Newsletter from XYZ Corp   Description: We hope you're enjoying our latest updates. Check out what's new this
month in your industry.
2      Subject: Your Invoice #123456 from ABC Corp Description: Please find your invoice for the recent services you availed from us.
Let us know if you have any questions.
3      Subject: Invitation to Join Webinar on Data Privacy   Description: Join us for a live session on upcoming data privacy
regulations and how your business can prepare.
4      Subject: Confirmation of Your Flight Booking           Description: Your flight from New York to London has been confirmed. Find
attached your e-ticket and itinerary.
5      Subject: Job Interview Scheduled for Tomorrow         Description: We are excited to meet you. Your interview is scheduled for
10:00 AM tomorrow. Best of luck!
6      Subject: You Won $1,000,000 - Claim Your Prize Now!   Description: Click here to claim your grand prize. This offer is only
valid for 24 hours!
7      Subject: Earn $500 Daily From Home Without Any Experience!   Description: Start earning today with our easy online job. No
skills required. Sign up now!
8      Subject: Urgent: Your Account Has Been Suspended       Description: We have detected suspicious activity in your account. Click
this link to resolve the issue immediately.
9      Subject: Limited Time Offer: Free Vacation to the Bahamas   Description: You've been selected for a free vacation! All you
need to do is cover the processing fee.
10     Subject: You Have Been Selected for a $100 Amazon Gift Card   Description: Congratulations! Click here to claim your free
Amazon gift card.
11     Subject: Project Update: Q4 Strategy Meeting           Description: Please find attached the latest updates from our Q4 planning
session. Let me know your thoughts.
12     Subject: Friendly Reminder: Team Lunch Tomorrow         Description: Just a reminder that we have our team lunch at 12 PM
tomorrow. Don't forget to RSVP!
13     Subject: Your Amazon Order Has Shipped                 Description: Your recent order has been shipped and is expected to arrive by Friday.
Track your package here.
14     Subject: Invitation: Networking Event in Your Area       Description: You're invited to a networking event with industry leaders.
Join us for an evening of insights and connections.
15     Subject: Weekly Status Update: Product Development       Description: Here's your weekly update on the product development
progress. We're on track for the Q1 launch.
16     Subject: Your Loan Has Been Pre-Approved               Description: You are eligible for a loan up to $100,000 with no credit check! Apply
now to get instant approval.
17     Subject: Immediate Action Required: Update Your Payment Information   Description: We couldn't process your recent
payment. Click here to update your payment details immediately.
18     Subject: Congratulations! You've Been Chosen for a Special Offer   Description: You've been selected for a limited-time offer.
Click now to avail this exclusive deal.
19     Subject: Get 100,000 Followers Instantly on Social Media   Description: Boost your social media presence with 100,000
followers. Click here to get started now!
20     Subject: Hot Singles in Your Area Looking to Chat         Description: Find singles near you! Join our platform today and start
connecting instantly.
21     Subject: Your Subscription Renewal Reminder             Description: Your subscription to our premium service is set to renew in 5 days. No
action is required unless you wish to cancel.
22     Subject: Meeting Confirmation: Marketing Strategy Discussion   Description: This is a confirmation for your meeting with the
marketing team on Monday at 2 PM.
23     Subject: Re: Follow-up on the Contract Details           Description: I'm following up on the contract we discussed last week. Let
me know if you have any further questions.
24     Subject: Welcome to the Team!                           Description: We are thrilled to have you join our company. Looking forward to seeing you at the
orientation next week.
25     Subject: Your Package is Out for Delivery               Description: Your order is on its way! Expect delivery by the end of the day.
26     Subject: Survey: Help Us Improve Our Services           Description: Please take 5 minutes to fill out this survey and help us
provide better services for you.
27     Subject: Invitation: Join Our Customer Success Webinar   Description: We'd like to invite you to a webinar focused on customer
success strategies for 2024.
28     Subject: Confirm Your Attendance at the Annual Gala       Description: We are excited to have you join us for our Annual Gala next
month. Please confirm your attendance.
29     Subject: Your Payment Has Been Received                 Description: Thank you for your payment. Your transaction has been successfully
processed.
30     Subject: Action Required: Update Your Account Password   Description: As part of our security measures, we require you to update
your password every 90 days.
31     Subject: Get Rid of Your Debt Now!                       Description: Struggling with debt? We can help you clear all your dues in no time!
Click here to get started.
32     Subject: Click Here for Free Gift Cards!                 Description: Get free gift cards from your favorite stores! Limited time offer,
claim yours now.
33     Subject: Your Account Is About to Be Closed             Description: Your account will be deactivated unless you confirm your details. Click
here to keep your account active.
34     Subject: You've Won a Brand New Car!                     Description: Congratulations! You've been selected to win a brand-new car. Just fill
out this form to claim your prize.
35     Subject: Earn Passive Income from Home                  Description: Start earning money with little effort. Join our program and make
thousands every week!
36     Subject: Last Chance to Claim Your Discount Voucher!     Description: This is your last chance to claim a 70% discount voucher.
Don't miss out!
```


37	Subject: Protect Your PC from Viruses Now	Description: Install our antivirus software and protect your PC from threats. Click here for a free trial.
38	Subject: Immediate Action Required: Resolve Payment Issue	Description: We were unable to process your recent payment. Click here to resolve the issue immediately.
39	Subject: New Hot Deal: Get 70% Off on Electronics	Description: Save big on the latest gadgets. Hurry, offer ends soon!
40	Subject: Congratulations! You've Been Approved for a Credit Card	Description: Apply now and get approved for a high-limit credit card with no annual fee.
41	Subject: Your Software License Has Been Renewed	Description: Your software license has been successfully renewed for the next 12 months.
42	Subject: Client Feedback Request for Last Project	Description: Could you please provide feedback on the recent project we completed? We value your input.
43	Subject: Travel Itinerary for Your Upcoming Trip	Description: Attached is the travel itinerary for your upcoming business trip. Let us know if you have any changes.
44	Subject: Notification: New Features Added to Your Account	Description: We've added exciting new features to your account. Log in to explore the updates.
45	Subject: Reminder: Payment Due in 3 Days	Description: Your payment is due in 3 days. Please ensure timely payment to avoid any service interruptions.
46	Subject: Weekly Newsletter: Industry Insights and News	Description: Here's this week's edition of our newsletter, filled with insights and news from your industry.
47	Subject: Annual Company Retreat Details	Description: We are pleased to share the details for our upcoming annual company retreat. Please review the attached agenda.
48	Subject: Job Offer: Congratulations, You're Hired!	Description: We are delighted to extend an offer for you to join our team. Please review the offer letter attached.
49	Subject: Appointment Confirmation: Dr. Smith at 10:30 AM	Description: This is a confirmation for your appointment with Dr. Smith tomorrow at 10:30 AM. Please arrive 15 minutes early.
50	Subject: Upcoming Webinar: Boosting Sales with AI	Description: Join us for a webinar on how to boost your sales using AI-powered tools and techniques.

About Author

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Senior Salesforce (SFDC), AI, and CRM Program Architect, highly skilled in integrating cutting-edge technologies like artificial intelligence and customer relationship management (CRM) platforms. With over **20 years of IT experience** (including 12 years in CRM/Salesforce and 5 years in AI), the author is a recognized leader in designing and delivering innovative AI and CRM solutions across industries. They have extensive expertise in **security, multi-org setups, data integration, design patterns, DevOps, and AI strategy**, and hold **20 Salesforce and 5 AI certifications**.

Feel free to reachout me for any AI/CRM free suggestions

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